



STANDARD COLLEGE NTUNGAMO

S.5 PHYSICS

END OF TERM TWO 2025

2hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

Answer **all** questions,

ITEM ONE

(a) William Thomson invented a galvanometer to detect weak signal current on long submarine telegraph cables. The galvanometer has a mirror attached to its coil. A light beam acting as a pointer is reflected from the mirror on to a scale 1m away, and a current causes 2mm deflection on the scale.

TASK: As a physics student

- (i) Derive the principle of operation of such an instrument.
 - (ii) Explain how the light beam helps in measurement of current in the instrument.
 - (iii) Determine how much deflection will be observed for the same current if the scale is moved away 2m to increase sensitivity.
- (b) A school carpenter fixed a dressing mirror in of the dormitories in a school. Irene the tallest girl among all who reside in that dormitory complained of not being able to see the whole of herself in the mirror while all others can comfortably see themselves. The carpenter has been advised to change the size of the mirror. Irene is 1.8m tall and her eyes are 10cm below the top of her head.

TASK: As a physics student,

- i) Advise the carpenter on the appropriate type of the mirror to use.
- ii) Help the carpenter to know the minimum height of the mirror to fix for Irene to see her full image.
- iii) Help the carpenter to know how close the lower edge of the mirror must be from the floor.

ITEM 2

A community health center in a remote area lacks digital thermometers. They instead use liquid-in-glass thermometers. During the malaria outbreak, many patients show high fever and accurate temperatures become crucial for treatment. At 0°C temperature, Boiling point and temperature θ of a patient, the length of the liquid column was 25.0cm, 37.5cm and 42.2cm respectively. There was also an argument that alcohol is a better liquid for use in thermometers compared to mercury.

Task: As a learner of physics,

- a) (i) State the thermometric property used by this thermometer
(ii) State the properties of a good thermometric property
- b) (i) Calculate the temperature of the patient
(ii) Explain why different thermometers based on different thermometric properties may give different values of temperature.
- c) Do you support the argument in the scenario? If yes, justify

ITEM 3

- a) A sound source on a car produces sound waves of frequency 850Hz. A stationary observer measures the emitted frequency using a detector which can measure a maximum frequency of 800Hz. He finds out that the detector only registers a reading while the car is moving.

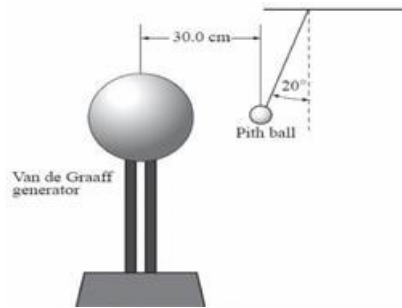
Hint: Speed of sound in air = 340ms^{-1}

Task: As a learner of physics,

- i) State and explain the effect in the above context
- ii) Explain whether the car must move towards or away from the observer for the detector to register a reading
- iii) Calculate the minimum speed at which the car must move for the detector to register the maximum reading
- iv) State the applications of the effect in (i) above

ITEM 4

In an electrostatic experiment to determine electrostatic force on a 2.5×10^{-4} Kg neutral pith ball, a student charges a Van de Graff generator with 2m diameter sphere holding a charge of $+5.0\mu\text{C}$ and suspends the pith ball by an insulating thread.



The pith ball gets attracted towards the van de Graaff generator at an angle of 20° to the vertical.

TASK: As a student of Physics,

- i) Explain how the van de Graaff generator acquires charge.
- ii) Explain why the pith ball gets attracted.
- iii) Find the tension in the thread and the magnitude of attractive force acting between the generator and the pith ball.
- iv) Find the potential on the surface of the sphere before the pith ball is suspended.
- v) Explain the effect of suspending the pith ball on the potential of the sphere.
- vi) How many electrons are detached off the sphere of the generator to acquire the charge?

ITEM 5

A body moving with a uniform acceleration covers a distance of 5.5 m in its 4th second and 9.5m in its 8th second of the motion. Also, two balls P of mass 8kg and Q of mass 5kg travelling in the same line in opposite directions with speeds 6ms^{-1} and 15ms^{-1} respectively make a perfect inelastic collision and momentum is conserved.

TASK

- (a) (i) Derive the third equation of motion and prove it is dimensionally consistent.
(ii) Find the initial velocity and the acceleration of the body.
- (b) Use Newtons laws of motion to show that when two bodies collide, their momentum is conserved.
- (c) (i) Find the velocity of P
(ii) Change in kinetic energy after collision between P and Q

END

